

Microcontroller 8051

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Microcontroller

A microcontroller also called MCU (Microcontroller Unit) is a small computer on a single metal-oxide-semiconductor (MOS) integrated circuit (IC) chip. A microcontroller contains one or more CPUs (processor cores) along with memory and programmable input/output peripherals. Program memory in the form of ferroelectric RAM, NOR flash ROM also often included on chip, as well as a small amount of RAM. Microcontrollers are designed for embedded applications, in contrast to the microprocessors used in personal computers or other general purpose applications consisting of various discrete chips.

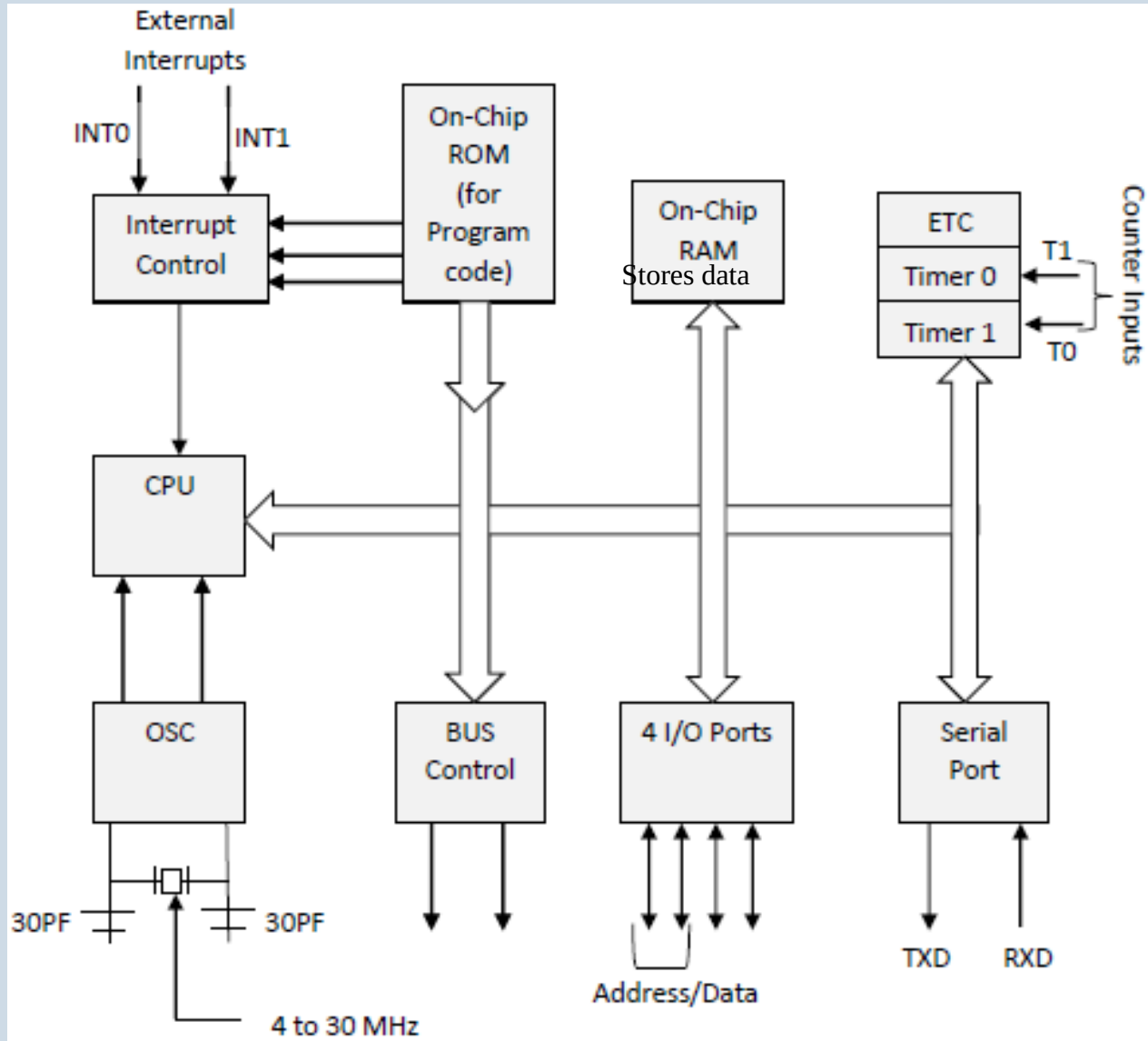


Figure 1: Architecture of Intel MCS-51 (commonly termed 8051).

The Intel MCS-51 (commonly termed 8051) is a single chip microcontroller (MCU) series developed by Intel in 1981 for use in embedded systems.

8051 microcontroller is designed by Intel in 1981. It is an 8-bit microcontroller. It is built with 40pins DIP (dual inline package).

❑ **Bus control-**

- Data bus-use to transfer 8 bit data.
- Address bus- use to transfer information.

❑ **Four general purpose parallel i/o port** –Each port is 8 bit port-

- Port 0 (P0):** Can be used as an 8-bit I/O port or as a multiplexed address/data bus (AD0-AD7) when accessing external memory.
- Port 1 (P1):** A general-purpose 8-bit I/O port.
- Port 2 (P2):** A general-purpose 8-bit I/O port.
- Port 3 (P3):** A general-purpose 8-bit I/O port, also used for various control signals like interrupt requests, timer outputs, and serial communication.

❑ **Timer and counter-** The 8051 microcontroller features two 16-bit timers/counters, Timer 0 (T0) and Timer 1 (T1), which can be used for timing events or counting external pulses, controlled by registers like TMOD and TCON.

❑ **Interrupt Control-** 2 way of giving interrupt to micro-controller

- By sending S/W instruction.
- By sending H/W signal.

❑ **Internal RAM and ROM-** 128 bytes of internal RAM (data memory) and 4KB of internal ROM (program memory).

Internal RAM (Data Memory):

- 128 bytes of on-chip RAM for temporary data storage.
- Used for storing data, intermediate results, and register banks.
- Includes register banks, read/write data, bit-addressable registers, and stack memory.
- Address range: 0000H to 007FH.

Internal ROM (Program Memory):

- 4KB of on-chip ROM for storing the microcontroller's program code.
- Contains data that are read-only.
- Address range: 0000H to FFFFH.

❑ **Serial port-** It consists of four parallel 8-bit ports, which are programmable as well as addressable as per the requirement and a serial port for transmitting and receiving data.

❑ **On-chip crystal oscillator (OSC)-** is integrated in the microcontroller having crystal frequency of 12 MHz.

❑ **CPU-** Microcontroller 8051 has a central processing unit which is also called ALU (Arithmetic Logic Unit) which performs all arithmetic and logical operation.

Application

- ✓ Light sensing and controlling devices like LED.
- ✓ Temperature sensing and controlling devices like microwave oven.
- ✓ Fire detection and safety devices like fire alarm.
- ✓ Measuring devices like volt meters.